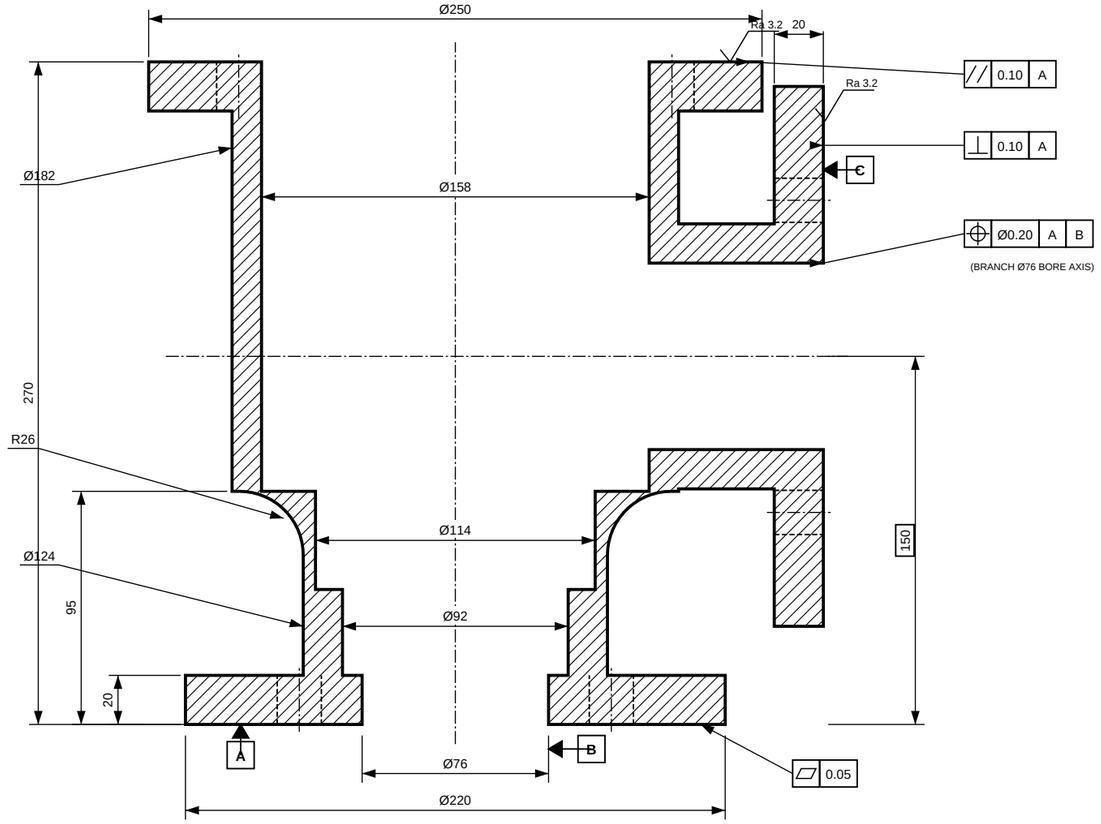
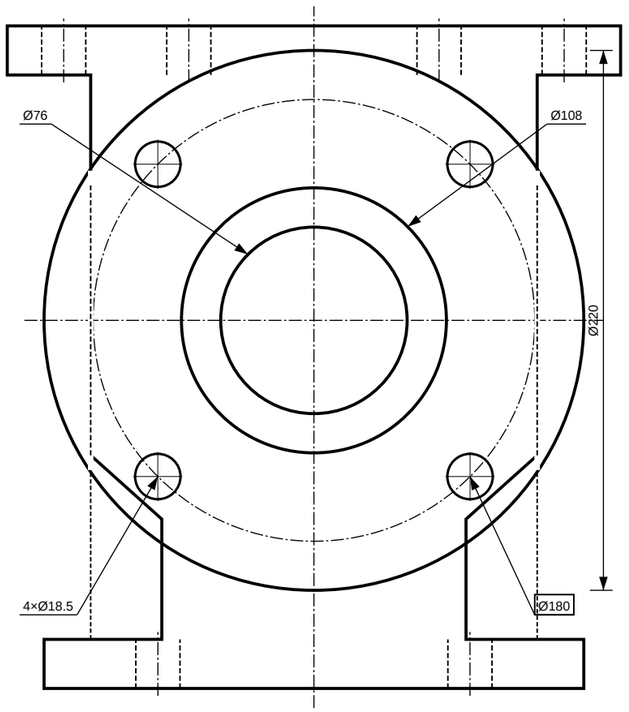
GENERAL: Ra 12.5

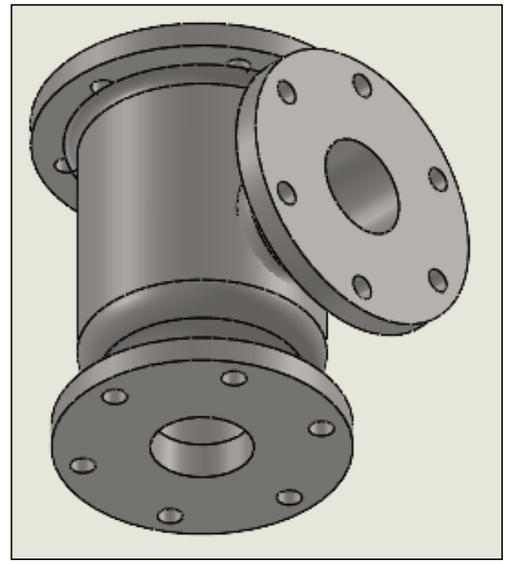
FRONT VIEW - SECTION A-A (SCALE 1:2.5)



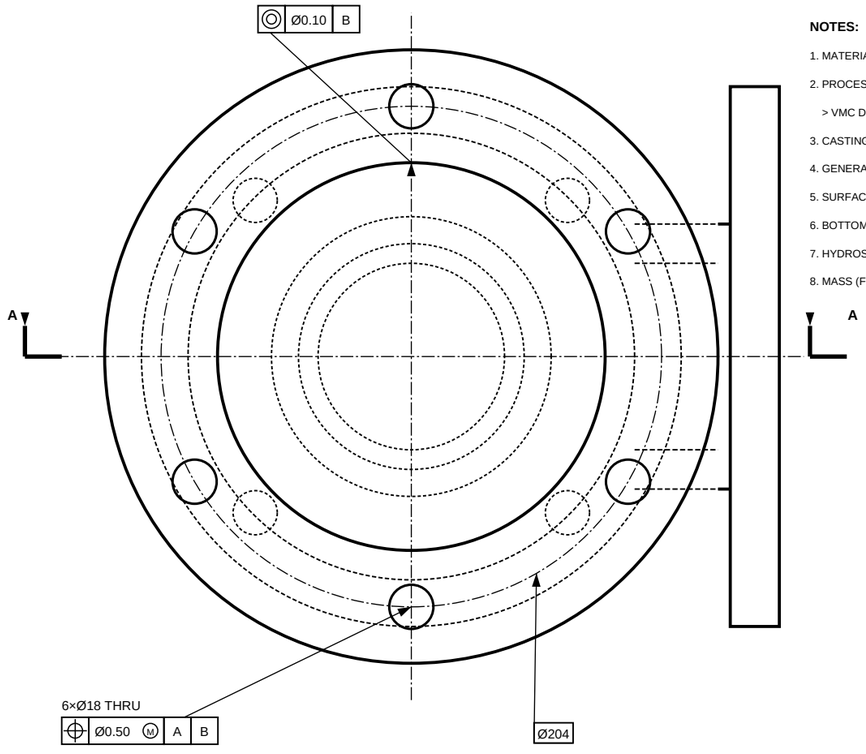
LEFT SIDE VIEW



ISOMETRIC VIEW (NOT TO SCALE)



TOP VIEW (SCALE 1:2.5)



NOTES:

1. MATERIAL: CAST CARBON STEEL ASTM A216 WCB (EQUIV. TO AISI 1020 IN MODEL). NORMALISE + STRESS RELIEVE AFTER CASTING.
2. PROCESS ROUTE: PATTERN & SAND CASTING > FETTLE > HEAT TREAT > CNC TURNING (FLANGE FACES, BORES, STEPS) > VMC DRILLING / SPOT-FACING OF BOLT HOLES > DEBURR > INSPECTION (CMM) > DPT ON MACHINED FACES.
3. CASTING TOLERANCE ISO 8062 CT10; DRAFT 2° ON CAST SURFACES; MACHINING ALLOWANCE 3 mm ON ALL MACHINED FACES.
4. GENERAL TOLERANCE ISO 2768-mK. FILLETS R3 AND EDGE BREAK 0.5×45° UNLESS STATED. R26 AS SHOWN.
5. SURFACE FINISH: FLANGE FACES Ra 3.2; Ø76 / Ø92 / Ø114 BORES Ra 1.6; ALL OTHER SURFACES AS CAST (Ra 12.5).
6. BOTTOM FLANGE: 4×Ø18 THRU ON PCD Ø180 (45° TO BRANCH AXIS). BRANCH FLANGE: 4×Ø18.5 ON PCD Ø180 AT 45°.
7. HYDROSTATIC TEST AND NDT AS PER CUSTOMER / APPLICABLE CODE. NO LEAKS OR POROSITY ON MACHINED FACES.
8. MASS (FINISHED, FROM CAD): 34.76 kg | APPROX. RAW CASTING: 37-38 kg. DO NOT SCALE DRAWING.

GD&T SUMMARY (ASME Y14.5-2018)

FEATURE	CONTROL
A: Bottom flange underside	Datum A (mounting face)
B: Ø76 main bore axis	Datum B
C: Branch flange face	Datum C (150 from B)
All 3 flange faces	Flatness 0.05
Top flange face	Parallel 0.10 | A
Branch flange face	Perp. 0.10 | A
Ø158 bore	Coaxial Ø0.10 | B
Branch Ø76 bore axis	Position Ø0.20 | A | B
6×Ø18 holes, PCD Ø204	Position Ø0.50 M | A | B
4×Ø18 holes, btm flange PCD Ø180	Position Ø0.50 M | A | B
4×Ø18.5 holes, branch PCD Ø180	Position Ø0.50 M | C | B

BASIC DIMENSIONS (BOXED): 150, Ø204, Ø180

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN MILLIMETRES
GENERAL TOL.: ISO 2768-mK
CASTING TOL.: ISO 8062 CT10
FINISH: Ra 12.5 AS CAST
LINEAR ±0.5 | ANGULAR ±1°

	NAME	DATE
DRAWN	AAKASH K. PANDEY	30-10-2026
CHECKED		
APPROVED		
MFG		
Q.A.		

MATERIAL: **CAST STEEL ASTM A216 WCB (≈ AISI 1020)**

HEAT TREATMENT / PROCESS: **NORMALISED; SAND CAST + CNC MACHINED**

WEIGHT (FINISHED): **34.76 kg**

SCALE: **1 : 2.5**

RAW CASTING (APPROX.): **≈ 37.5 kg**

ORGANISATION: **YUVAINTERN - MECHANICAL DESIGN INTERNSHIP**

TITLE: **FLANGED ANGLE BODY (90° BRANCH, 3 FLANGES)**

DRAWING NO.: **id0.8**

SIZE: **A3**

REV.: **A**

SHEET: **1 OF 1**

PROJECTION: FIRST ANGLE